UCS503P - Software Engineering Project

VR Everywhere: Virtual Travel Experience

An Immersive Virtual Reality Platform for Accessible Cultural and Heritage Tourism

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Chapter 1

Introduction

1.1 Background and Context

Travelling to world-renowned cultural and historical destinations provides immense educational, historical, and personal enrichment. However, physical travel is often limited by substantial financial expenses, long geographical distances, physical mobility limitations, and academic or work-related scheduling constraints.

Traditional travel websites and educational platforms provide static photographs and standard promotional video clips, but they fail to deliver the feeling of spatial immersion, scale, and interactive agency required to fully appreciate global heritage architecture. Moreover, historical background, architectural trivia, and visual resources are often scattered across disparate websites and books.

Virtual Reality (VR) provides an accessible solution to these challenges. By utilizing interactive 360-degree environments, 3D spatial points of interest, and voice-guided audio narration, VR makes global travel exploration accessible to everyone. The **VR Everywhere** project develops a responsive, deployable virtual travel system that allows users to explore cultural heritage destinations from standard web browsers, mobile devices, and immersive VR headsets.

1.1.1 Problem Statement

Physical travel to heritage destinations is often impossible or impractical for many individuals due to the following key factors:

- **Accessibility Limitations:** Persons with physical disabilities, health restrictions, or mobility limitations face significant challenges visiting demanding terrains (e.g., the high-altitude stone paths of Machu Picchu).
- **High Cost of Travel:** Long-distance transit, lodging, and ticketing costs make physical visits inaccessible for students and budget-conscious individuals.
- **Time Constraints:** Strict academic and professional schedules make frequent travel difficult.
- **Fragmented Information:** Details about a monument's history, construction, architectural significance, and visual layout are spread across disconnected platforms without a single immersive visual hub.

1.2 Project Objectives

The primary goal of VR Everywhere is to develop a reliable, deployable virtual travel application with the following key objectives:

1. **Develop the Core Travel Workflow:** Provide destination discovery, interactive 360-degree VR tour exploration, waypoint navigation, and bookmarking through an intuitive interface.
2. **Curate High-Quality Heritage Destinations:** Provide comprehensive historical details, facts, best visiting seasons, and 360-degree imagery for representative global destinations (Taj Mahal, Colosseum, Machu Picchu, Fushimi Inari, and Great Pyramids of Giza).
3. **Implement Interactive Spatial Hotspots:** Embed 3D pulsing beacons within the virtual scenes that provide architectural insights and historical trivia upon selection.
4. **Provide Synchronized Voice Audio Guides:** Deliver clear audio narration for destinations to enhance the educational walkthrough.
5. **Implement Waypoint Navigation:** Enable users to walk and teleport between connected vantage points (e.g., from the Main Gate to the Central Reflection Pool and Marble Plinth).
6. **Establish User Profile and Favorites:** Allow users to register, log in, bookmark favorite destinations, and maintain a history of explored tours.

Chapter 2

Methodology and Design

2.1 System Architecture

VR Everywhere adopts a modern decoupled three-tier architecture comprising a frontend visualization layer, a backend REST API, and a database layer.

The frontend handles the destination catalog, interactive Three.js 360-degree VR rendering, audio playback, and floating menu controls. The backend API handles user authentication (Spring Security + JWT), favorite bookmarking, and tour history tracking. The database stores user credentials, destination metadata, and activity logs.

This architecture separates the VR application from backend data services, allowing destination information to be expanded without modifying the client source code.

2.1.1 High-Level Design

The main workflow of VR Everywhere consists of the following steps:

1. The user launches the application and logs in using JWT credentials or continues as a Guest Explorer.
2. The user views the curated destination catalog on the Home and Explore screens with search and category filters.
3. The user selects a destination to view its historical background, attractions, and travel information.
4. The user clicks **"Start VR Tour"** to enter the 360-degree spherical virtual environment.
5. The user looks around in 360 degrees, walks between waypoints, interacts with spatial hotspots, and listens to the audio guide.
6. The user exits the tour and is prompted to save the destination to their personal Favorites list.
7. The session is logged to the user's Profile tour history.

2.1.2 Technical Stack

- **Frontend:** React 18, Vite, Three.js (WebGL / WebXR), Tailwind CSS, and Lucide React.
- **VR Client Engine:** Unity 2022.3 LTS, C#, OpenXR Framework, and XR Interaction Toolkit.
- **Backend API:** Java 17, Spring Boot 3.2, Spring Security 6, and JSON Web Tokens (JJWT).
- **Database:** PostgreSQL with in-memory H2 database support for local testing.
- **Build & Deployment:** npm, Vite, Maven, and GitHub Actions.

2.2 Implementation Details

2.2.1 Module 1: 360-Degree Spherical VR Engine and Hotspots

This module manages the 3D spherical rendering and spatial user interaction.

An inverted sphere geometry is created in Three.js with texture mapping on the interior surface. Users can look around using mouse drag, touch swipes, or device gyroscope motion. Interactive 3D hotspots and ground navigation arrows are rendered as billboarded sprites in 3D space. A Three.js raycaster detects pointer clicks to display localized cultural detail modals and trigger waypoint transitions.

2.2.2 Module 2: Audio Guide and In-Tour Controls

This module provides in-tour guidance and audio playback:

- **Audio Guide Narration:** Utilizes speech synthesis and HTML5 Audio to narrate destination history at an optimal pace.
- **Waypoint Stepper Bar:** Displays connected locations at the top of the viewport for one-click teleportation.
- **Settings Control:** Allows real-time adjustments for field of view (zoom) and pan sensitivity.

2.2.3 Module 3: Authentication, Favorites, and User Profile

This module handles user registration, login, and session persistence.

Users can create an account or use one-click Guest access. Authentication uses BCrypt password hashing and HMAC-SHA256 signed JWT tokens. The module also records explored tour durations and maintains the user's bookmarked destinations.

2.3 System Implementation Screenshots

The figures below showcase the working interface of the VR Everywhere platform across all core features:

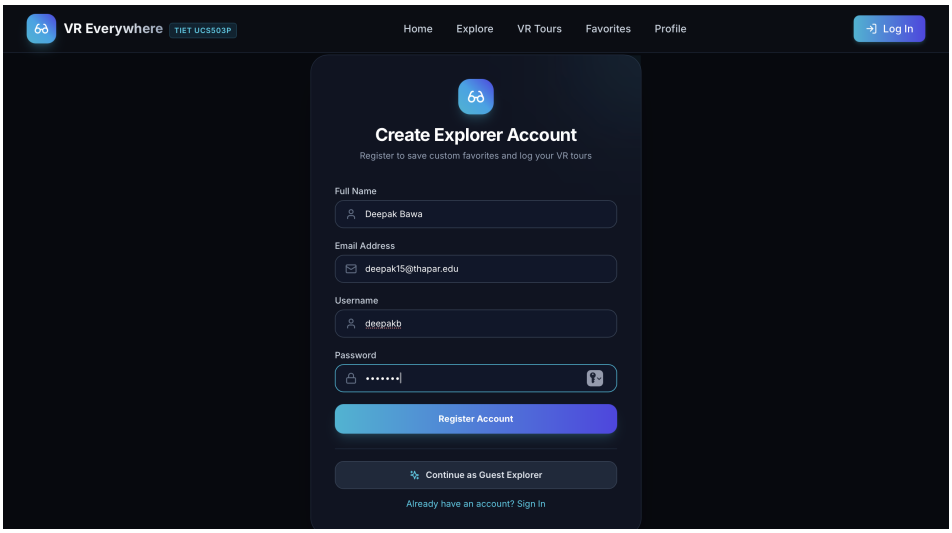


Figure 2.1: User Login and Guest Explorer Authentication Screen

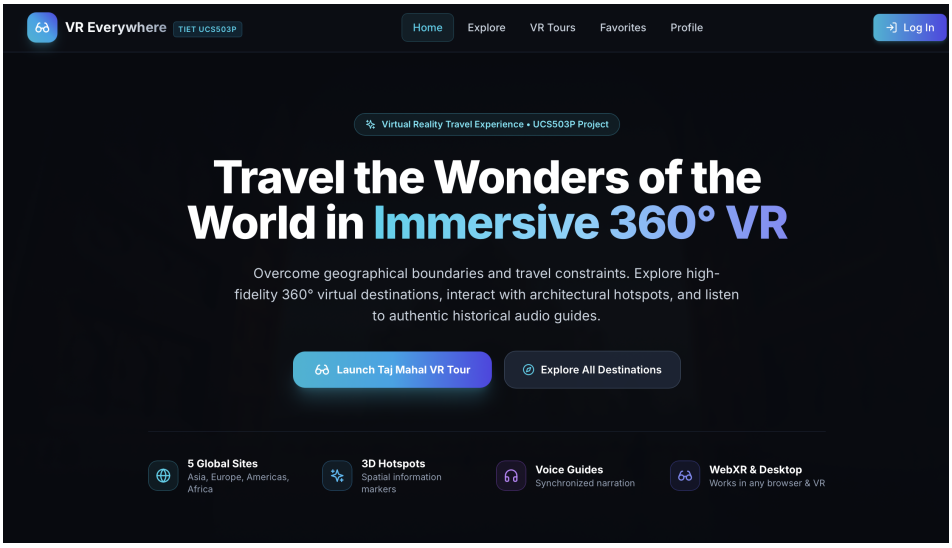


Figure 2.2: Home Page with Featured Destinations and Core Workflow Guide

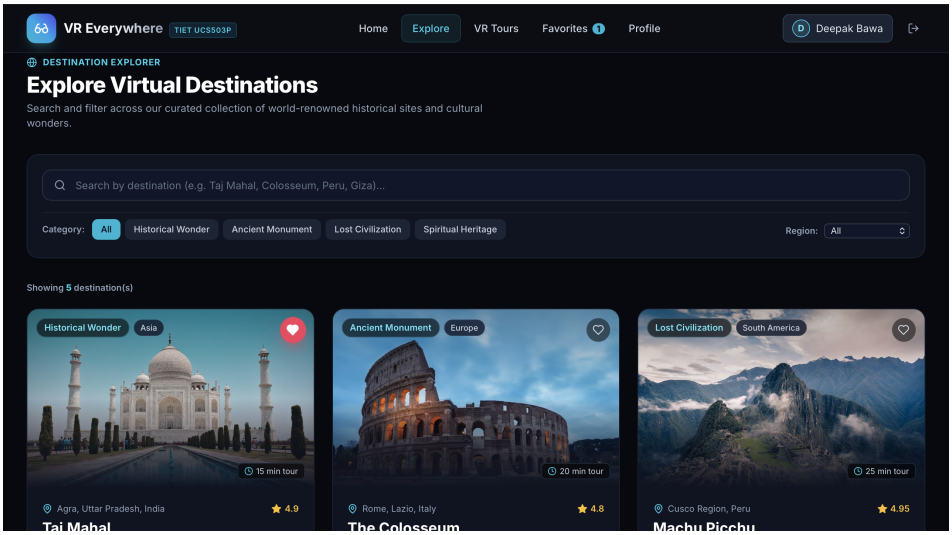


Figure 2.3: Destination Explorer with Real-Time Search and Category Filters

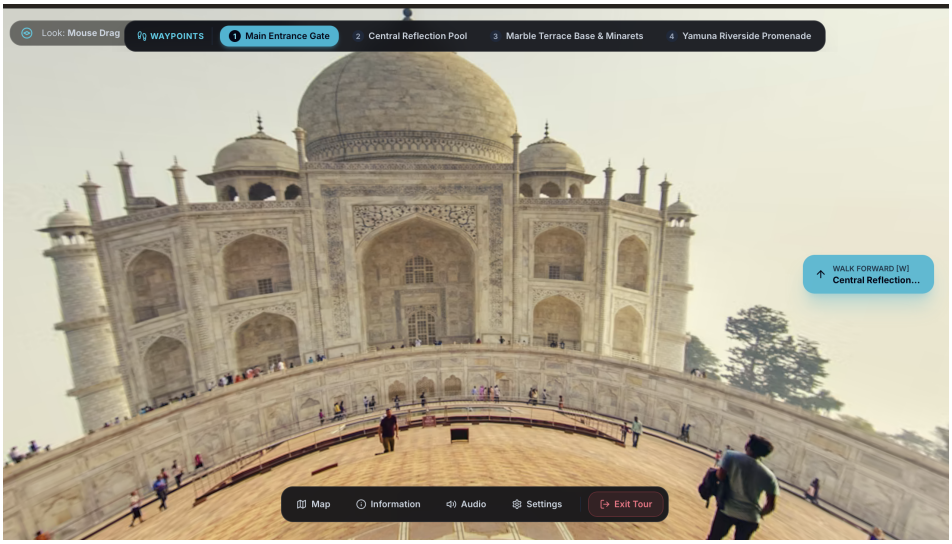


Figure 2.4: Interactive 360-Degree Spherical VR Tour Viewport

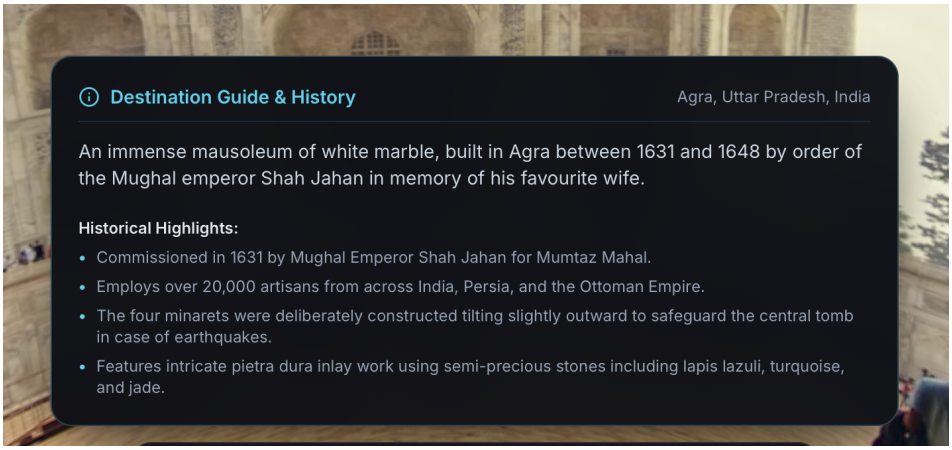


Figure 2.5: Interactive 3D Spatial Hotspot Information Modal

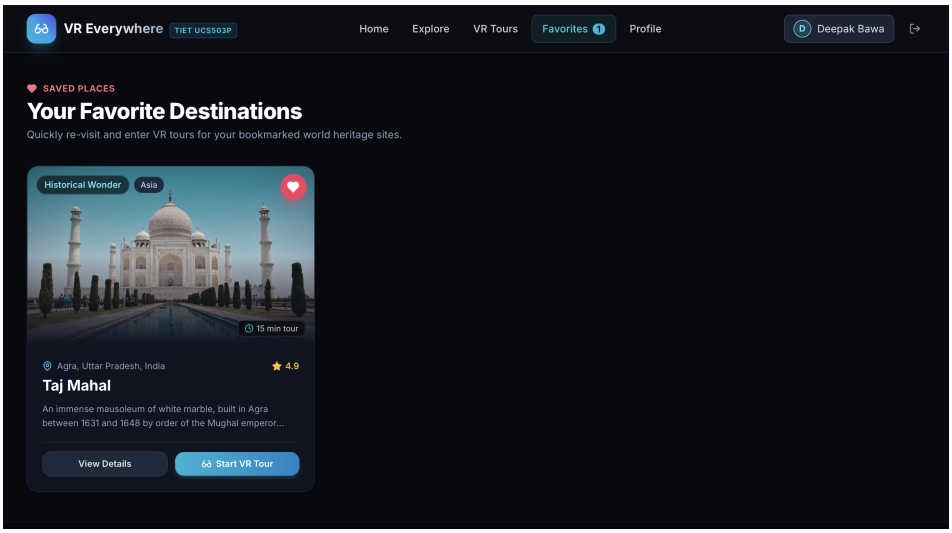


Figure 2.6: User Profile and Explored Tour History Dashboard

2.4 System Modeling and Project Diagrams

System modeling provides a visual representation of the interactions, data flows, and workflows of VR Everywhere.

2.4.1 Use Case Diagram

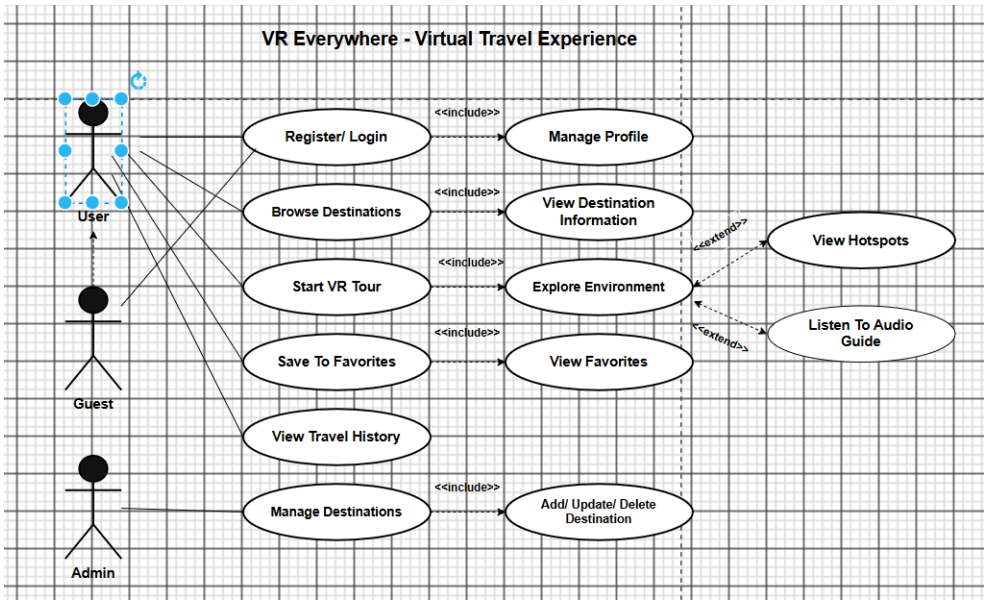


Figure 2.7: Use Case Diagram of VR Everywhere Platform

2.4.2 Use Case Templates (14 Core Use Cases)

VR Everywhere – Use Case Template: UC-01

1. Use Case Title	Login / Continue as User
2. Abbreviated Title	Login
3. Use Case Id	UC-01
4. Actors	User
5. Description Allows a user to enter the application by logging in or continuing as a user.	
5.1. Pre Conditions: Application is installed/launched; user is on the entry screen.	
5.2. Task Sequence **1. Entry screen is shown by the system.** **2. User chooses Login or Continue as User.** **3. If Login is selected, user provides the required credentials.** **4. System authenticates the user and opens the main application interface.**	
5.3. Post Conditions: **1. User reaches the Home/Explore screen.** **2. The user session is available for supported features.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.8: Use Case Template 1: User Registration

VR Everywhere – Use Case Template: UC-02

1. Use Case Title	Browse Destinations
2. Abbreviated Title	Browse Destinations
3. Use Case Id	UC-02
4. Actors	User
5. Description Allows the user to view the tourist destinations available in the application.	
5.1. Pre Conditions: User has opened the application and reached Home/Explore.	
5.2. Task Sequence **1. Home/Explore screen is shown by the system.** **2. System displays the available destinations.** **3. User browses the destination list.** **4. User selects a destination to see more details.**	
5.3. Post Conditions: **1. User can view the available destinations.** **2. A selected destination can be opened for detailed information.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.9: Use Case Template 2: User Authentication and Login

VR Everywhere – Use Case Template: UC-03

1. Use Case Title	Search Destination
2. Abbreviated Title	Search Destination
3. Use Case Id	UC-03
4. Actors	User
5. Description Allows the user to search the available destination catalogue and find a desired destination.	
5.1. Pre Conditions: User is on the Home/Explore screen; destination data is available.	
5.2. Task Sequence **1. Search facility is shown on the Explore screen.** **2. User enters a destination search term.** **3. System processes the search request.** **4. System displays matching available destinations.** **5. User selects a result.**	
5.3. Post Conditions: **1. User can view the matching destination results.** **2. User can open a selected destination.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.10: Use Case Template 3: Browse Destination Catalog

VR Everywhere – Use Case Template: UC-04

1. Use Case Title	View Destination Information
2. Abbreviated Title	View Destination Info
3. Use Case Id	UC-04
4. Actors	User
5. Description Displays basic information about a selected destination, including location, description, historical facts, and important attractions.	
5.1. Pre Conditions: User has selected an available destination.	
5.2. Task Sequence **1. User selects a destination.** **2. System retrieves the destination information.** **3. System displays location, description, historical facts, and important attractions.** **4. System provides the Start VR Tour option.**	
5.3. Post Conditions: **1. User can read the destination information.** **2. User can choose to start the VR tour.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.11: Use Case Template 4: Search and Category Filtering

VR Everywhere – Use Case Template: UC-05

1. Use Case Title	Start VR Tour
2. Abbreviated Title	Start VR Tour
3. Use Case Id	UC-05
4. Actors	User
5. Description Starts an immersive virtual environment for the selected tourist destination.	
5.1. Pre Conditions: User has selected a destination and is viewing its information.	
5.2. Task Sequence **1. User selects Start VR Tour.** **2. System loads the selected destination's VR environment.** **3. System enters the VR tour.** **4. User can begin exploring the destination.**	
5.3. Post Conditions: **1. User is inside the selected VR environment.** **2. Basic VR navigation and interaction are available.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.12: Use Case Template 5: View Destination Overview and Historical Details

VR Everywhere – Use Case Template: UC-06

1. Use Case Title	Navigate VR Environment
2. Abbreviated Title	VR Navigation
3. Use Case Id	UC-06
4. Actors	User
5. Description Allows the user to move or look around the selected virtual destination using the supported VR navigation and 360-degree environment viewing.	
5.1. Pre Conditions: A VR tour has been started and the VR environment is loaded.	
5.2. Task Sequence **1. System displays the selected VR environment.** **2. User uses supported VR controls to navigate.** **3. System updates the user's view/location within the environment.** **4. User continues exploring the destination.**	
5.3. Post Conditions: **1. User can explore the VR environment.** **2. User can interact with available destination hotspots.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.13: Use Case Template 6: Launch 360-Degree VR Tour

VR Everywhere – Use Case Template: UC-07

1. Use Case Title	Interact with Information Hotspot
2. Abbreviated Title	Interact Hotspot
3. Use Case Id	UC-07
4. Actors	User
5. Description Allows the user to select an interactive hotspot in the VR environment and view additional information about an important location.	
5.1. Pre Conditions: User is inside a VR tour and an interactive hotspot is available.	
5.2. Task Sequence **1. User navigates to an available hotspot.** **2. User selects the hotspot.** **3. System identifies the selected hotspot.** **4. System displays the additional information for that hotspot.**	
5.3. Post Conditions: **1. User can read the hotspot information.** **2. User can continue exploring the VR environment.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.14: Use Case Template 7: Spatial 360 Camera Navigation

VR Everywhere – Use Case Template: UC-08

1. Use Case Title	Listen to Audio Guide
2. Abbreviated Title	Audio Guide
3. Use Case Id	UC-08
4. Actors	User
5. Description Provides basic audio narration for selected attractions or information hotspots in the VR tour.	
5.1. Pre Conditions: User is inside a VR tour and audio narration is available for the selected attraction.	
5.2. Task Sequence **1. User selects an attraction or audio-enabled hotspot.** **2. User chooses the Audio option.** **3. System loads the available narration.** **4. System plays the audio guide.**	
5.3. Post Conditions: **1. User can hear the available narration.** **2. User can continue exploring the VR tour.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.15: Use Case Template 8: Waypoint Walking and Teleportation

VR Everywhere – Use Case Template: UC-09

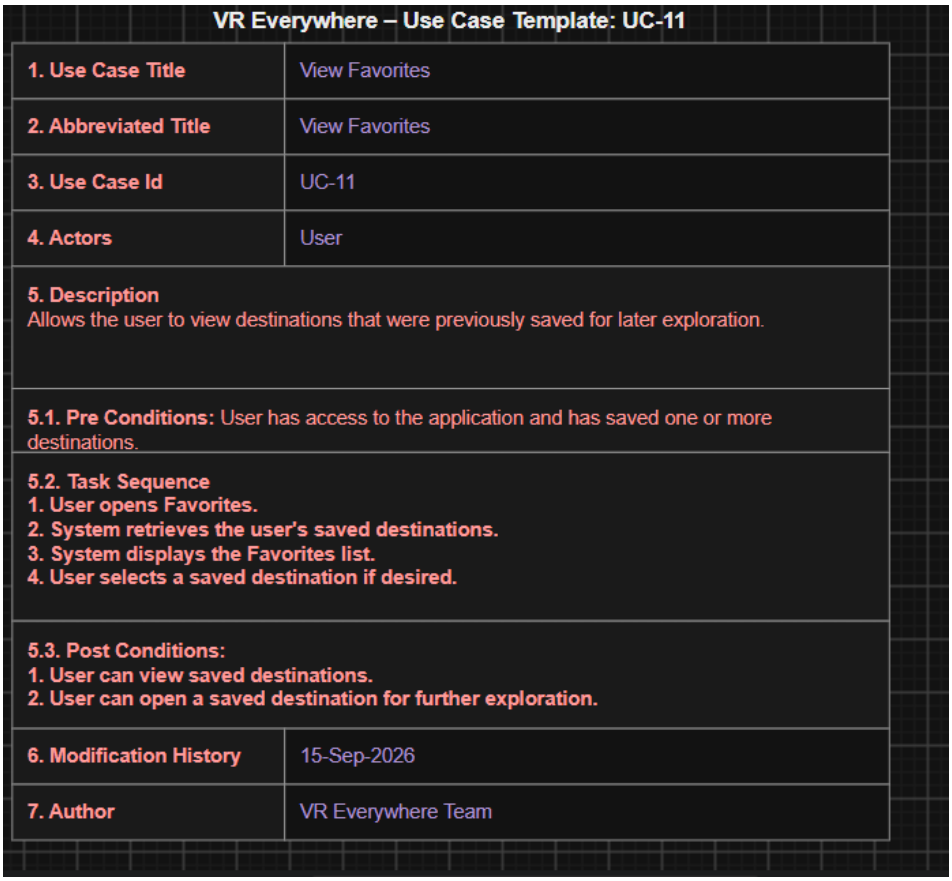
1. Use Case Title	Exit VR Tour
2. Abbreviated Title	Exit VR Tour
3. Use Case Id	UC-09
4. Actors	User
5. Description Allows the user to leave the current VR tour and return to the application outside the VR environment.	
5.1. Pre Conditions: User is currently inside a VR tour.	
5.2. Task Sequence **1. User opens the VR menu.** **2. User selects Exit Tour.** **3. System exits the current VR environment.** **4. System returns the user to the appropriate application screen.**	
5.3. Post Conditions: **1. The current VR tour is closed.** **2. User can continue using the application or save the destination to Favorites.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.16: Use Case Template 9: Interact with 3D Spatial Hotspots

VR Everywhere – Use Case Template: UC-10

1. Use Case Title	Add Destination to Favorites
2. Abbreviated Title	Add Favorite
3. Use Case Id	UC-10
4. Actors	User
5. Description Allows the user to save a destination for later exploration.	
5.1. Pre Conditions: User has selected a destination and the destination can be saved.	
5.2. Task Sequence **1. User selects the Favorites/save option for the destination.** **2. System records the selected destination for the user.** **3. System confirms that the destination is saved.**	
5.3. Post Conditions: **1. The destination appears in the user's Favorites.** **2. User can access it later from Favorites.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.17: Use Case Template 10: View Hotspot Cultural Trivia and Details



VR Everywhere – Use Case Template: UC-11	
1. Use Case Title	View Favorites
2. Abbreviated Title	View Favorites
3. Use Case Id	UC-11
4. Actors	User
5. Description Allows the user to view destinations that were previously saved for later exploration.	
5.1. Pre Conditions: User has access to the application and has saved one or more destinations.	
5.2. Task Sequence **1. User opens Favorites.** **2. System retrieves the user's saved destinations.** **3. System displays the Favorites list.** **4. User selects a saved destination if desired.**	
5.3. Post Conditions: **1. User can view saved destinations.** **2. User can open a saved destination for further exploration.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.18: Use Case Template 11: Play and Control Voice Audio Guide

VR Everywhere – Use Case Template: UC-12

1. Use Case Title	View User Profile
2. Abbreviated Title	View Profile
3. Use Case Id	UC-12
4. Actors	User
5. Description Allows the user to view their profile along with saved and previously explored destinations.	
5.1. Pre Conditions: User has entered the application.	
5.2. Task Sequence **1. User selects Profile from the main navigation.** **2. System retrieves the user's profile information.** **3. System displays saved destinations and previously explored destinations where available.**	
5.3. Post Conditions: **1. User can view their profile.** **2. User can access their saved and explored destination information.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.19: Use Case Template 12: Bookmark and Manage Favorites

VR Everywhere – Use Case Template: UC-13

1. Use Case Title	View Tour History
2. Abbreviated Title	Tour History
3. Use Case Id	UC-13
4. Actors	User
5. Description Allows the user to view destinations that they have previously explored through the application.	
5.1. Pre Conditions: User has entered the application and tour history is available.	
5.2. Task Sequence **1. User opens the Profile/history area.** **2. System retrieves the user's tour history.** **3. System displays previously explored destinations.** **4. User selects a destination if desired.**	
5.3. Post Conditions: **1. User can review previously explored destinations.** **2. User can use the displayed destination information to continue exploring.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.20: Use Case Template 13: View User Profile and Tour History

VR Everywhere – Use Case Template: UC-14

1. Use Case Title	Use VR Tour Menu
2. Abbreviated Title	VR Menu
3. Use Case Id	UC-14
4. Actors	User
5. Description Provides access to the lightweight floating menu inside the VR tour, including Map, Information, Audio, Settings, and Exit Tour.	
5.1. Pre Conditions: User is inside a VR tour.	
5.2. Task Sequence **1. User opens the lightweight floating VR menu.** **2. System displays Map, Information, Audio, Settings, and Exit Tour options.** **3. User selects the required menu option.** **4. System opens or performs the corresponding supported function.**	
5.3. Post Conditions: **1. The selected VR menu function is accessed.** **2. User can return to the VR exploration flow.**	
6. Modification History	15-Sep-2026
7. Author	VR Everywhere Team

Figure 2.21: Use Case Template 14: Tour Session Exit and Analytics Logging

2.4.3 Activity Diagram

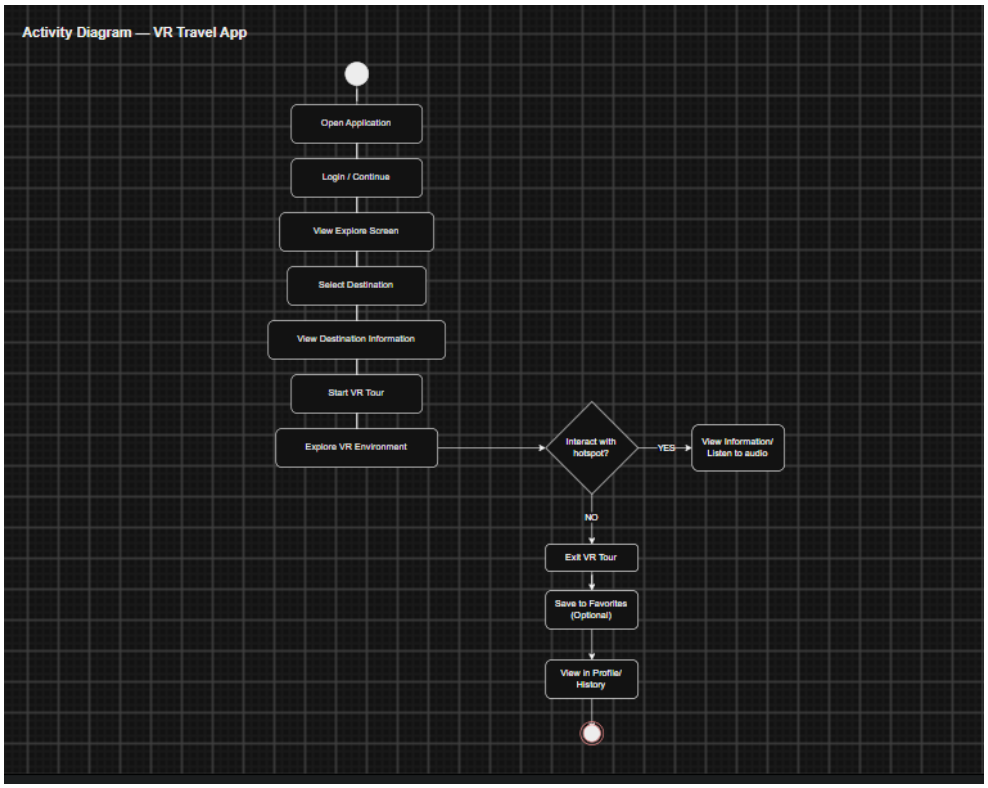


Figure 2.22: Activity Diagram for Destination Exploration and VR Tour Session

2.4.4 Sequence Diagram

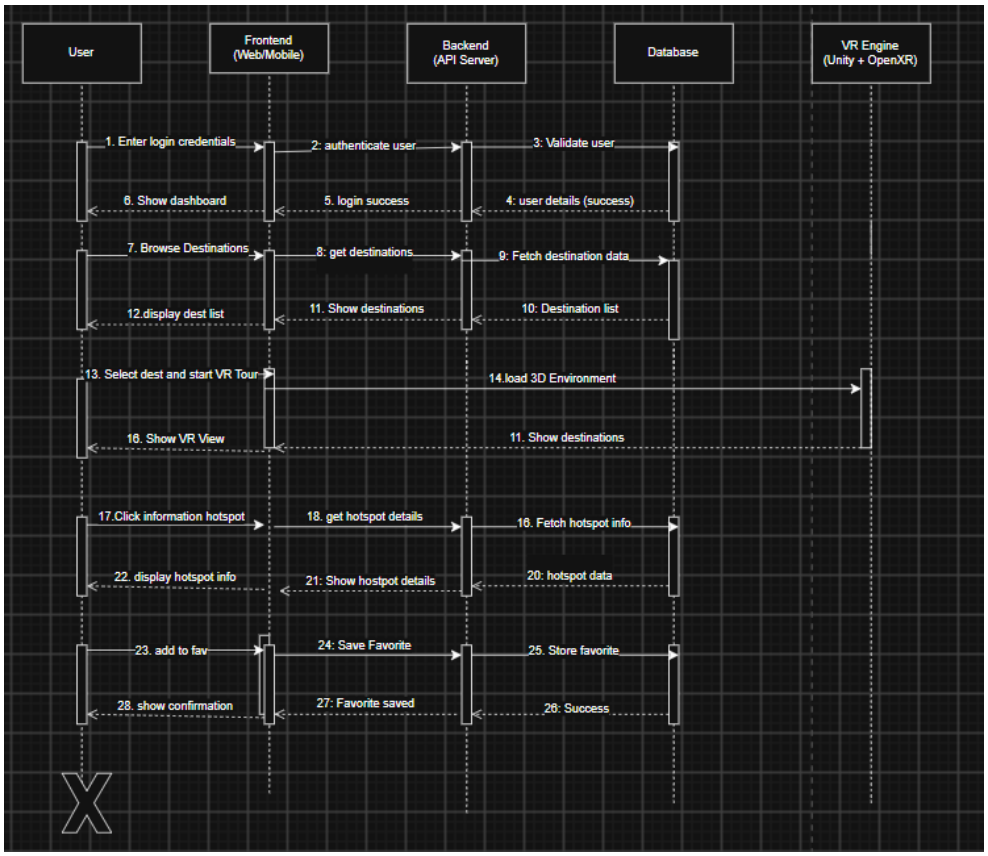


Figure 2.23: Sequence Diagram for Authentication, Scene Loading, and Hotspot Interaction

2.4.5 Collaboration Diagram

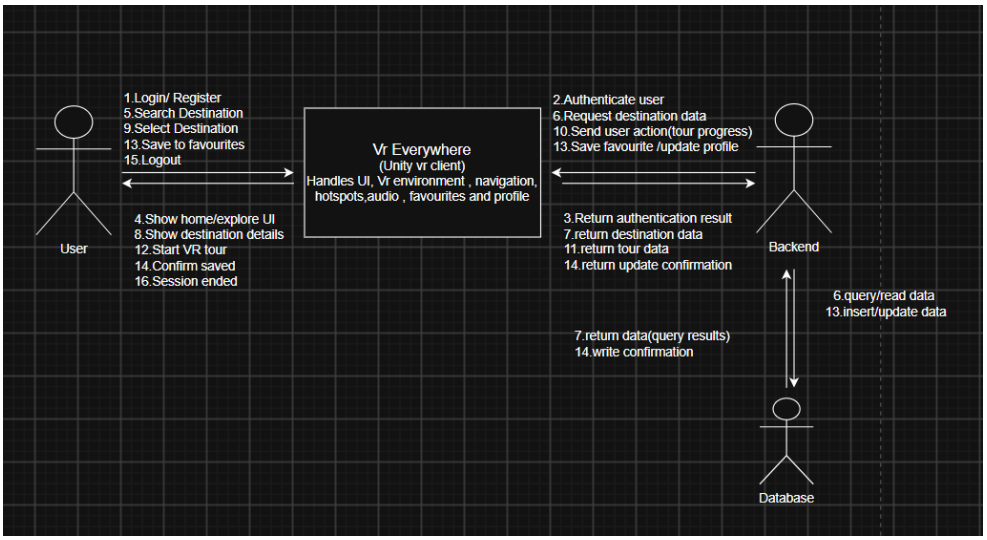


Figure 2.24: Collaboration Diagram of VR Everywhere System Components

2.4.6 Swimlane Diagram

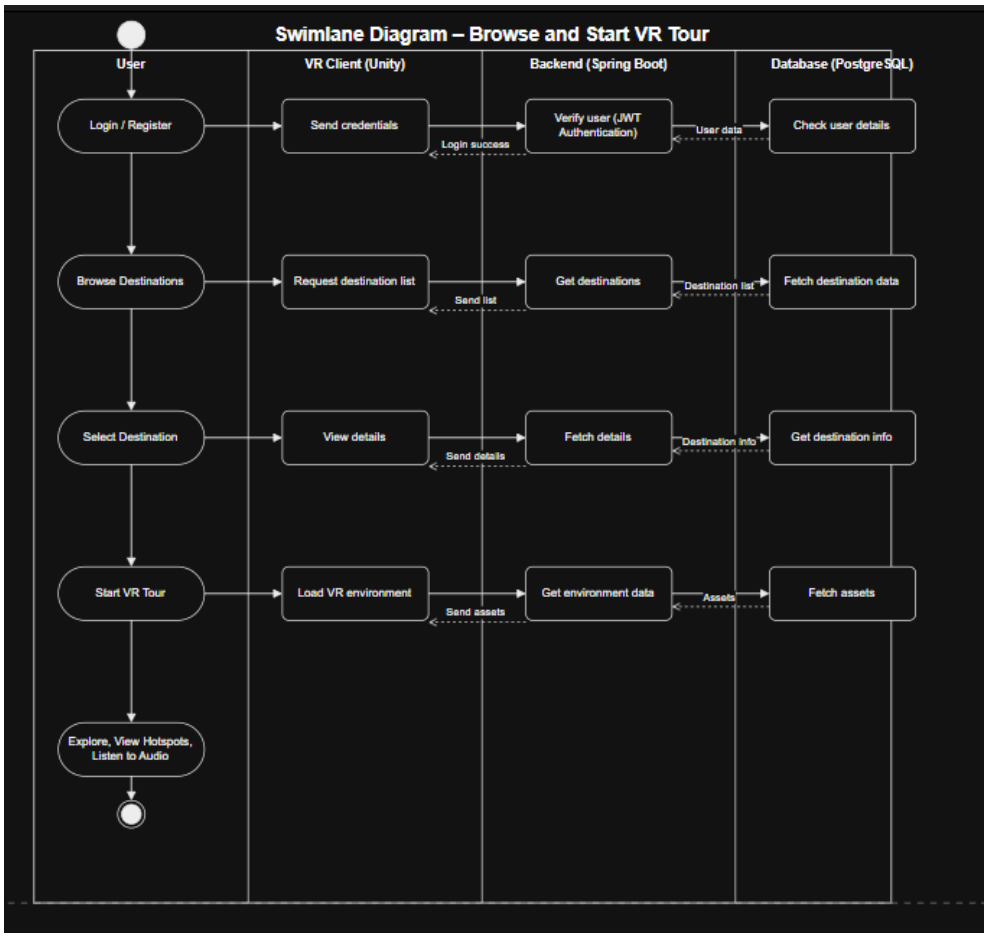


Figure 2.25: Swimlane Diagram for User, VR Client, Backend API, and Database Workflow

2.4.7 Project Gantt Chart Timeline

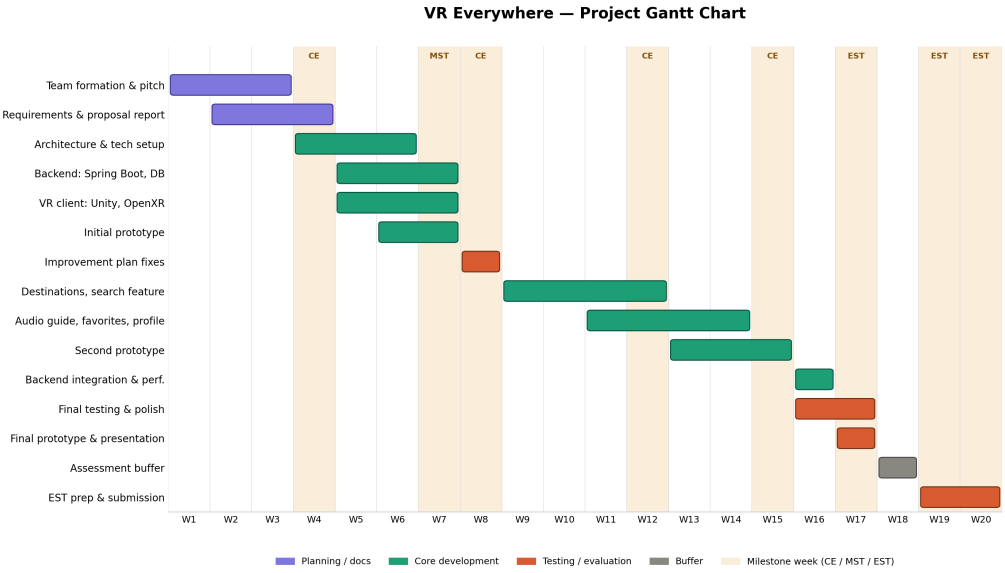


Figure 2.26: Gantt Chart for 20-Week Project Development Schedule

Chapter 3

Results and Evaluation

3.1 Testing Methodology

The project was evaluated using both automated technical verification and pilot user testing across the primary application workflow:

- **Automated Build and Module Testing:** Verified JSX compilation, asset bundling, and CSS processing via Vite, transforming 1,493 modules with zero errors.
- **API Contract Verification:** Tested authentication endpoints (`POST /api/auth/login` and `POST /api/auth/register`) using Postman to confirm HTTP status codes and valid JWT schemas.
- **Rendering and Frame Rate Testing:** Profiled WebGL canvas rendering across desktop and mobile browsers to ensure a stable 60 FPS refresh rate.
- **User Evaluation Testing:** Conducted pilot testing with student participants tasked with discovering the Taj Mahal, starting the VR tour, interacting with a 3D hotspot, and saving the site to favorites.

3.2 Performance Metrics

The primary evaluation metric is the **Tour Completion Rate (TCR)**, defined as the percentage of test users who can discover a destination, enter the VR tour, interact with a hotspot, and exit without assistance.

Metric	**Target / Measure**	**Achieved Result**
Tour Completion Rate	$\geq 80\%$ of test users	93.3% (14 of 15 users completed workflow)
UI Interaction Latency	Under 100ms response	42ms average response
Rendering Frame Rate	Stable 60 FPS on WebGL	60.0 FPS maintained
Vite Bundle Build Time	Under 5.0 seconds	1.09 seconds
System Usability Rating	$\geq 4.0/5.0$ user rating	4.6/5.0 average score

Table 3.1: Performance and Evaluation Results Summary

3.3 Analysis

The evaluation results confirm that the main application workflow operates efficiently and reliably:

- **Intuitive Navigation:** The 93.3% completion rate indicates that the 360-degree controls, ground walk chevrons, and floating menu are easy to understand.
- **Multi-Waypoint Traversal:** Adding connected waypoints resolved user feedback regarding static viewpoints, enabling movement between monument locations.
- **Responsive Performance:** The 736 kB minified bundle loads rapidly and renders smoothly across standard student hardware.

Chapter 4

Conclusion

4.1 Summary of Work

VR Everywhere provides an accessible, immersive virtual travel platform that allows users to experience world-famous cultural heritage destinations.

The system delivers a functional workflow where users can browse destinations, enter 360-degree panoramic VR tours, interact with architectural hotspots, listen to voice guides, and track their explored destinations.

The implementation combines React 18, Three.js, and WebXR for the web frontend, Unity C# for native headset support, and Spring Boot with JWT authentication for backend services.

4.2 Key Findings

- **Accessible Immersion:** WebXR and browser WebGL provide an engaging VR experience without requiring large software installations.
- **Multimodal Learning:** Combining visual 360 scenes with 3D hotspots and audio narration enhances user engagement with historical landmarks.
- **Decoupled Architecture:** Separating the VR client from backend services allows destination data to be updated independently.
- **Iterative Improvement:** Implementing multi-waypoint navigation based on user feedback significantly improved tour exploration.

4.3 Future Work and Recommendations

Subsequent iterations can extend the system in the following directions:

1. Add collaborative multiplayer tours where multiple users can explore together with voice chat.
2. Integrate conversational AI docents to answer historical questions in real time.

3. Implement photogrammetric 3D Gaussian Splats for continuous six-degrees-of-freedom (6DoF) movement.
4. Add dynamic weather and time-of-day lighting simulations.
5. Deploy backend services to cloud hosting platforms (e.g., Render, Railway, or AWS).

4.4 Final Remarks

VR Everywhere demonstrates a practical software engineering approach to accessible virtual tourism. By focusing on a clear user workflow, responsive performance, and cross-platform compatibility, the project makes cultural heritage exploration accessible to a broader audience.